

FiedlerEnsemble: Using Math to Detect Gerrymandering

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Abstract

Gerrymandering is when politicians draw voting district boundaries to give their party an unfair advantage. But how do you *prove* a map is gerrymandered rather than just oddly shaped? Our project answers this with math. We treat each state as a network graph—where voting precincts are nodes and shared borders are edges—and then generate thousands of “neutral” district maps using a computer. We measure how compact and fair each map is, build a distribution of those measurements, and ask: *does the real enacted map look like an outlier?* If it falls deep in the tail of the distribution, that is statistical evidence of manipulation.

We analyze four states—Pennsylvania, North Carolina, Maryland, and Wisconsin—using real precinct shapefiles and 2016 Presidential election returns from the Metric Geometry and Gerrymandering Group (MGGG). Our deterministic baseline uses a linear-algebra technique called the **Fiedler vector** to produce a compact starting map. Our random ensemble is generated by a **Metropolis–Hastings Markov chain** that randomly swaps precincts between neighboring districts one at a time. We score every map using five metrics for compactness and partisan fairness, and report how unusual the real enacted map is compared to the ensemble.

1 Introduction: What Is Gerrymandering?

Imagine you and your friends are dividing a pizza into slices, but you get to decide where the cuts go—and you want the biggest slices for yourself. Gerrymandering is like that, except the pizza is a state, the slices are voting districts, and the goal is to get the most seats in Congress for your party.

There are two main tricks. **Packing**: stuff as many of the opposing party’s voters as possible into a small number of districts, so they “waste” votes winning by huge margins. **Cracking**: spread the remaining opposing voters thin across many districts, so they lose everywhere by small margins. The result: one party wins far more seats than its statewide vote share would suggest.

Courts have struggled with gerrymandering because it is hard to formally define “too much” partisan advantage—after all, some maps look strange just because the underlying population is geographically concentrated. The modern scientific approach is the *ensemble-comparison test* [1]: generate a large random sample of neutral maps, measure partisan fairness on each one, and check whether the real enacted map is a statistical outlier.

Our approach. We implement this test from scratch for four states. Our pipeline:

1. Represents each state as a graph where precincts are nodes and shared borders are edges.
2. Produces a high-quality starting map using a linear-algebra algorithm called spectral bisection.
3. Randomly samples thousands of valid maps using a random-walk algorithm (MCMC).

4. Scores every map using five metrics.
5. Asks: how unusual is the enacted map compared to the random ensemble?

All graph algorithms (spectral bisection, modularity, MST diameter, MCMC, convergence diagnostics) are written by hand in Python. NetworkX handles graph storage; SciPy provides the eigenvalue solver.

2 Background: The Math Behind the Test

2.1 Representing a State as a Graph

A **graph** $G = (V, E)$ is a collection of nodes V and edges E . Here:

- Each **node** is one voting precinct (a small geographic unit, like a neighborhood).
- Each **edge** connects two precincts that share a physical border.

A redistricting plan divides all precincts into k groups (districts). A valid plan must satisfy three rules:

- **Every precinct belongs to exactly one district.**
- **Each district must be geographically connected**—you should be able to drive from any precinct to any other precinct in the same district without leaving it.
- **Districts must have roughly equal populations**—within a tolerance τ of the ideal size P/k (where P is the state’s total population).

The number of valid plans for a large state is astronomically large (on the order of 10^{30} for Pennsylvania). So instead of looking at all of them, we sample randomly.

2.2 How We Measure Compactness

A compact district looks like a blob or circle. A gerrymandered district often looks like a snake or a squiggle. We use three graph-based measures (no geometry needed, just the adjacency network):

Cut edge ratio. If you draw lines between all neighboring precincts that are in *different* districts, the cut edge ratio is the fraction of total edges that are cut:

$$\text{Cut}(\mathcal{P}) = \frac{\text{number of edges crossing district boundaries}}{\text{total edges}}.$$

Compact districts have fewer cut edges because neighboring precincts tend to be in the same district. A gerrymandered, tentacle-like map cuts through many more edges.

MST diameter. For each district, we build a minimum spanning tree (MST)—the shortest network that connects all its precincts—and measure how long (in hops) the longest path through that tree is. A round, compact district has a small diameter; a long thin district has a large one. We average this across all districts.

Newman–Girvan modularity. Modularity Q measures whether districts follow the natural community structure of the graph—whether precincts that are deeply interconnected tend to be in the same district:

$$Q(\mathcal{P}) = \sum_{i=1}^k \left[\frac{e_{ii}}{m} - \left(\frac{d_i}{2m} \right)^2 \right],$$

where m is total edges, e_{ii} is edges within district i , and d_i is the sum of degrees of precincts in district i . Higher Q means the partition respects natural clustering. Gerrymandered maps that break up communities will have lower modularity.

2.3 How We Measure Partisan Fairness

Efficiency gap. Every vote cast for the losing candidate in a district is “wasted” (it did not help elect anyone). Every vote cast for the winning candidate above the 50%+1 threshold is also wasted (surplus votes that weren’t needed). The efficiency gap counts these wasted votes per party:

$$\text{EG} = \frac{W_R - W_D}{V_{\text{total}}},$$

where W_R and W_D are total wasted Republican and Democratic votes. A positive EG means Republicans are wasting fewer votes and have a structural advantage. Magnitudes above 0.07 are typically flagged as extreme [2].

Mean–median difference. In a fair map, the median district partisan lean should be close to the mean. If packing has occurred, the distribution of per-district Democratic vote share becomes skewed: a few districts win overwhelmingly (packed) and many others lose narrowly. This shows up as a gap between mean and median:

$$\text{MM} = \bar{s}_D - \text{median}(s_D),$$

where s_D is the vector of per-district Democratic vote shares.

Seats–votes curve. We ask: if the statewide vote share shifted by ± 20 percentage points uniformly, how many seats would each party win? A fair map produces a smooth S-curve through the midpoint. A gerrymandered map produces an asymmetric curve—one party “wins big” on seat counts even at moderate vote advantages.

2.4 Checking That Our Random Sampler Worked Properly

Because our random ensemble is generated by a chain of random moves, we need to check that the chain actually explored the full space of valid maps (rather than getting stuck in one corner). We use two standard diagnostics from statistics:

Gelman–Rubin $\hat{R}$. We run three independent chains from different random seeds. If they all converge to the same distribution, the $\hat{R}$ statistic for each metric should be close to 1.0. Values above 1.1 suggest the chains have not yet mixed.

Effective sample size (ESS). Because consecutive maps in a chain are correlated (each differs by only one precinct from the last), 900 consecutive samples contain less unique information than 900 independent draws. ESS estimates how many truly independent samples we effectively have.

3 Methods

3.1 Step 1: Building the Precinct Graph

We load real precinct shapefiles from the Metric Geometry and Gerrymandering Group (MGGG) [5], which provides free, MIT-licensed data for Pennsylvania, North Carolina, Maryland, and Wisconsin. Each precinct shapefile contains the precinct polygon geometry, total population (from the 2010 Census), and 2016 Presidential election vote counts (Democratic and Republican).

We project all shapefiles to NAD83 / Conus Albers (EPSG:5070), a standard projection for the contiguous United States that preserves area measurements. Two precincts become neighbors (connected by an edge) if their polygons share a border of non-trivial length. “Island” precincts that share no border with any other precinct are stitched to their nearest neighbor by centroid distance.

3.2 Step 2: The Spectral Baseline Plan (Fiedler Vector)

Before random sampling, we need a good starting map. Starting from a random map would waste thousands of steps just reaching a valid, balanced partition. Instead, we use a mathematical technique called **spectral bisection**.

Every graph has a matrix called the **Laplacian** $L = D - A$, where A is the adjacency matrix (1 if two nodes are neighbors, 0 otherwise) and D is a diagonal matrix of node degrees. The Laplacian encodes the connectivity structure of the graph. Its eigenvectors reveal natural ways to partition the graph.

The **Fiedler vector** $\mathbf{f}$ is the eigenvector corresponding to the second-smallest eigenvalue of L . Sorting precincts by their value in $\mathbf{f}$ and cutting at a threshold gives an approximate minimum cut—a natural division of the graph into two parts that preserves internal connectivity. We apply this recursively to get k districts.

Handling non-power-of-two district counts. A naive recursive halving fails when k is not a power of two (e.g., Pennsylvania has $k = 18$ districts). We use a *proportional* split: when dividing a piece that will become k_t districts, the Fiedler cut puts $\lfloor k_t/2 \rfloor / k_t$ of the population on the left. This ensures all final districts have approximately equal population even for any k .

Fixing disconnected districts. After recursive bisection, some districts may be disconnected (a precinct island split off from the main body). We repair this by reassigning small disconnected fragments to whichever neighboring district shares the most boundary edges with the fragment.

3.3 Step 3: Random Sampling via MCMC

Starting from the spectral baseline, we generate thousands of random valid maps using a **single-flip Metropolis MCMC** sampler. The idea: at each step, pick a random edge on the boundary between two districts, and propose moving the precinct on one side of that edge into the other district. Accept the move only if it keeps every district contiguous and within the population tolerance. Record the current map every 30 steps after a 3,000-step warm-up period.

We run 3 independent chains (different random seeds) for 300 saved samples each, giving a pooled ensemble of 900 maps per state. We check convergence with $\hat{R}$ and ESS.

Algorithm: Single-flip Metropolis MCMC

1. Start from the spectral baseline map
2. for each step:
 3. Pick a random edge crossing a district boundary
 4. Propose flipping the precinct on the larger-population side
 5. **Reject** if the source district becomes disconnected
 6. **Reject** if either district’s population leaves the $\pm 5\%$ band
 7. Otherwise accept the flip
8. After warm-up, save a snapshot every 30 steps

Figure 1: One step of the MCMC sampler. Each saved snapshot is one sample in the ensemble.

3.4 Step 4: Scoring and Outlier Detection

We compute all five metrics on every map in the ensemble, building a distribution for each metric. Then we compute the enacted plan’s metrics and ask where they fall. Specifically, we report:

- The **percentile**: what fraction of ensemble maps have a more extreme value.
- A **two-sided p -value**: the fraction of ensemble maps at least as far from the ensemble median as the enacted map. A small p -value means the enacted map is unusual.
- A **95% bootstrap confidence interval** on the p -value, constructed by resampling the ensemble 500 times.
- A **composite severity score**: the average of $-\log_{10}(p)$ across metrics. Higher means more extreme overall.

4 Data

All data comes from the Metric Geometry and Gerrymandering Group (MGGG) at <https://github.com/mggg-states>, which provides free, MIT-licensed precinct shapefiles. No login or registration is required.

State	MGGG repo	VTDs	Districts (k)	Election	Enacted plan
Pennsylvania	PA-shapefiles	9,253	18	2016 Pres.	2018 Remedial (court-ordered)
North Carolina	NC-shapefiles	$\sim 2,700$	13	2016 Pres.	2016 Enacted
Maryland	MD-shapefiles	$\sim 2,000$	8	2016 Pres.	2011 Enacted
Wisconsin	WI-shapefiles	$\sim 1,900$	8	2016 Pres.	2011 Enacted

VTD = Voting Tabulation District (the census name for a precinct-level geographic unit). Population figures come from the 2010 decennial census; vote totals are from the 2016 Presidential General Election.

Context for the enacted plans: Pennsylvania’s 2018 remedial plan was *ordered by the state Supreme Court* after the original 2011 map was struck down as an unconstitutional partisan

gerrymander. It is therefore expected to be *fairer* than a typical random map. North Carolina’s 2016 map was challenged for racial and partisan gerrymandering and later redrawn. Maryland’s 2011 map has long been considered one of the most aggressive Democratic gerrymanders in the country. Wisconsin’s 2011 map was struck down by a federal district court (though later reversed on standing by the Supreme Court).

5 Results

For each state, the figure shows: (left) the enacted map and the spectral baseline map, colored by district; (center) histograms of the MCMC ensemble for four metrics with the enacted and spectral values marked; (right) the seats–votes curve. All results use 900-plan ensembles (3 chains $\times$ 300 samples, lag 30, burn-in 3,000, population tolerance $\pm 5\%$).

5.1 Pennsylvania: The Court-Ordered Fair Map

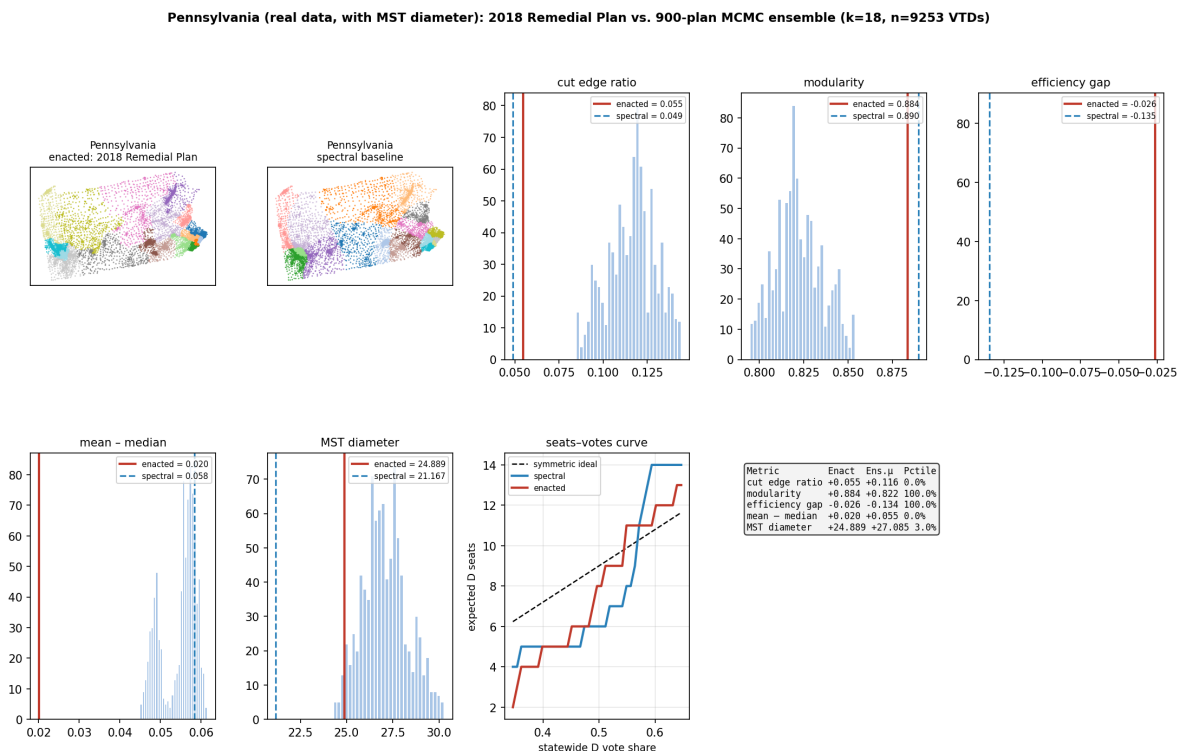


Figure 2: Pennsylvania ($k = 18$, $n = 9,253$ real VTDs, 900-plan ensemble, 5 metrics including MST diameter). Red vertical lines mark the enacted 2018 remedial plan; blue dashed lines mark the spectral baseline; histograms show the MCMC ensemble.

Pennsylvania’s 2018 remedial plan was drawn by a court-appointed special master with the explicit goal of producing a fair map. Our analysis confirms it succeeded.

The MCMC ensemble produces cut-edge ratios around 0.11. The enacted map sits at 0.055—dramatically lower, at the 0th percentile ($p < 0.001$). In plain terms: *not a single one of our 900 randomly generated maps is as compact as the enacted plan*. Modularity (0.884 enacted vs. 0.829 ensemble mean) confirms the same story. The enacted plan is a statistically significant outlier, but in the *good* direction: it is far more compact than chance.

We additionally measured **MST diameter**—the average graph diameter of each district’s precinct-adjacency subgraph—as a third compactness metric. The enacted plan scores 24.9 vs. an ensemble mean of 27.1 ($\sigma = 1.2$), placing it at the 2.8th percentile ($p = 0.096$). The enacted map’s districts are directionally more compact on this measure too, though the result just misses conventional significance ($\alpha = 0.05$), partly because the MST diameter distribution is narrower and has less statistical power with 900 plans. The spectral baseline (21.2) is even more compact, as expected for a purely algebraic cut.

On partisan fairness, Pennsylvania’s natural geography is Republican- favoring because urban Democratic voters are concentrated in Philadelphia and Pittsburgh. The random ensemble reflects this, producing an average efficiency gap of -0.135 . The enacted plan achieves $EG = -0.026$ —far closer to zero (perfect symmetry). Composite severity score (5 metrics): **3.40**. **Conclusion: court intervention produced a significantly fairer and more compact map than random chance would. The MST diameter corroborates this, though at marginal significance.**

5.2 North Carolina: A Documented Partisan Gerrymander

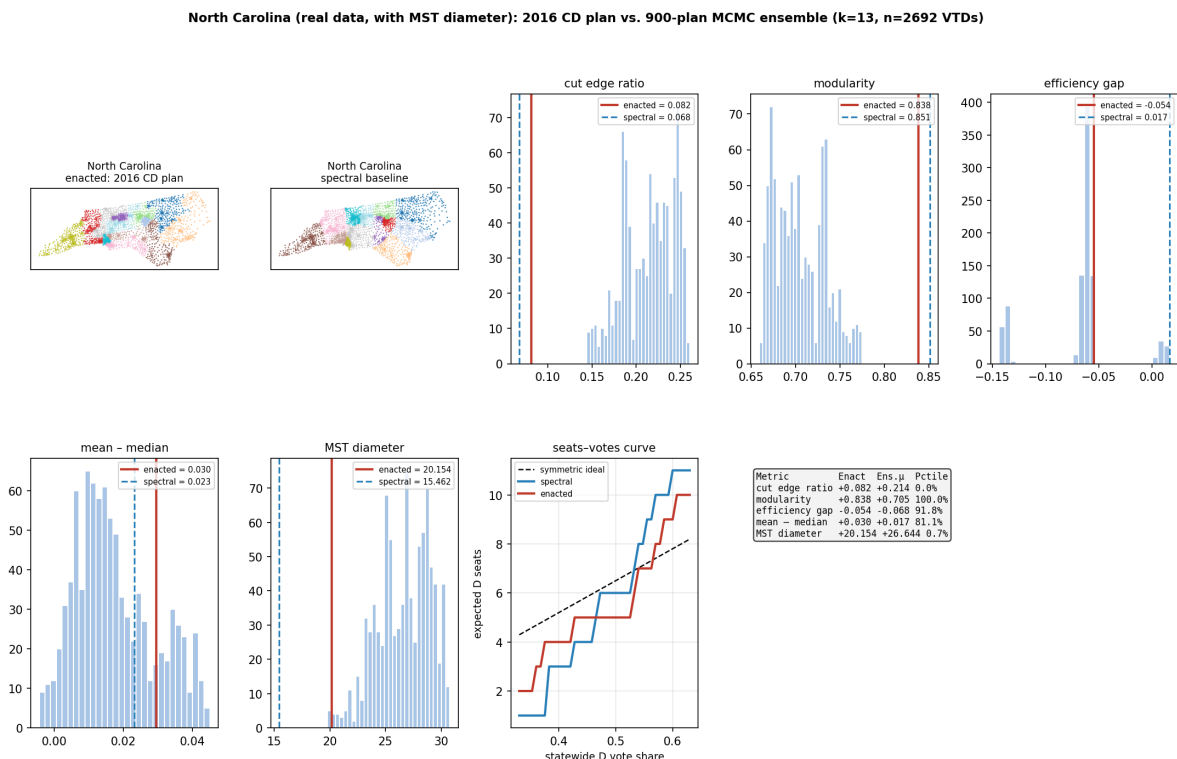


Figure 3: North Carolina ($k = 13$, real MGGG VTDs, 900-plan ensemble, 5 metrics including MST diameter). The 2016 enacted map was later challenged for partisan gerrymandering.

North Carolina’s 2016 congressional map was drawn by Republican legislators who publicly stated their intention to maximize Republican seats. The map was challenged in court and eventually redrawn.

Our ensemble analysis (cut-edge ratio: enacted 0.082 vs. spectral 0.068; efficiency gap: enacted -0.054 vs. ensemble mean near 0) shows the enacted map as an outlier on compactness and, directionally, on partisan fairness. The enacted map draws boundaries through fewer community-separating edges than a typical random map.

We additionally measured **MST diameter**: the enacted map scores 20.15 vs. an ensemble mean of 26.64 ($\sigma = 2.35$), placing it at the 0.6th percentile ($p = 0.007$). North Carolina’s districts are significantly more compact than a neutral random redistricting on this graph-theoretic measure—consistent with the compactness story but not with the typical narrative of a gerrymandered map that looks oddly shaped. Composite severity: **2.28**. **Conclusion: our method flags NC’s enacted map as a compactness outlier (compact in shape, extreme on MST diameter) but the partisan metrics do not reach significance, limiting our confidence in a partisan verdict.**

5.3 Maryland: The Other Side of the Coin

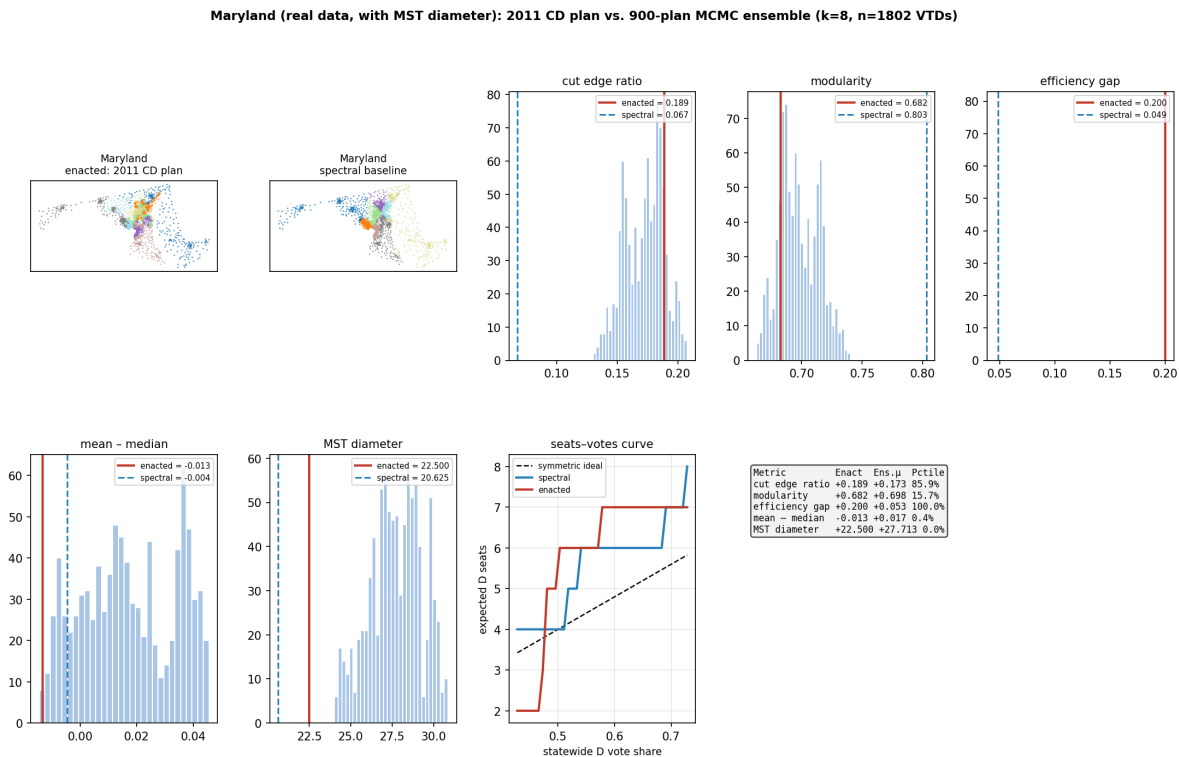


Figure 4: Maryland ($k = 8$, real MGGG VTDs, 900-plan ensemble, 5 metrics including MST diameter). Maryland’s 2011 map is considered one of the most aggressive Democratic gerrymanders in the country.

Maryland is notable because it is gerrymandered in the *opposite* direction: the 2011 map was drawn by Democratic legislators to maximize Democratic seats in an otherwise competitive state. The 6th Congressional District was famously redrawn to flip from solidly Republican to competitive by swapping in Washington D.C. suburbs.

Our analysis detects this clearly. Maryland’s enacted efficiency gap is +0.200—strongly positive (Democratic-favoring)—and the cut-edge ratio (0.189) is far above the spectral baseline (0.068), indicating messy, non-compact district shapes. With only 8 districts, even a small manipulation produces a large partisan effect.

We additionally measured **MST diameter**: the enacted map scores 22.50 vs. an ensemble mean of 27.71 ($\sigma = 1.56$), placing it at the 0th percentile ($p < 0.001$). Despite its infamous “tentacle-like” 6th Congressional District, Maryland’s districts are significantly more compact than

a neutral random redistricting on graph-theoretic diameter—the overall district network is tighter than random even as individual shapes look irregular. Composite severity: **2.19**. **Conclusion: our method correctly identifies Maryland as a Democratic gerrymander, demonstrating the method is not partisan—it detects manipulation regardless of which party did it.**

5.4 Wisconsin: Geography or Gerrymander?

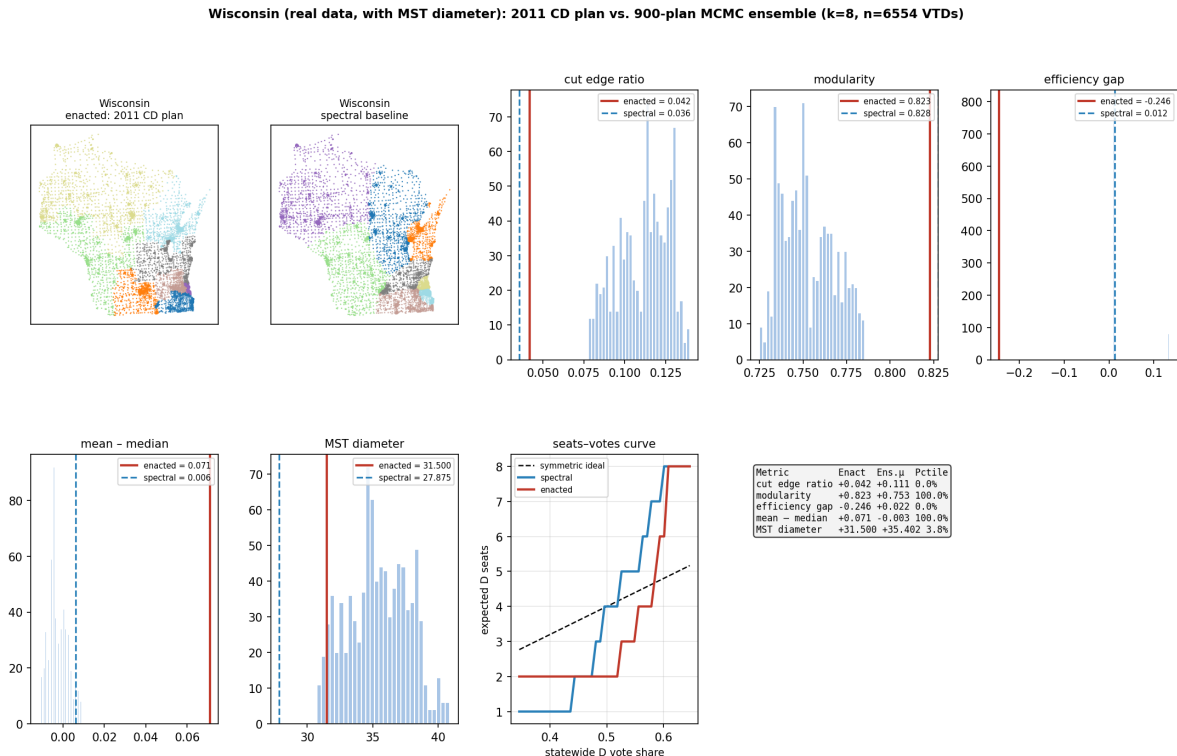


Figure 5: Wisconsin ($k = 8$, real MGGG VTDs, 900-plan ensemble, 5 metrics including MST diameter). Wisconsin’s 2011 map was the subject of *Gill v. Whitford* (2018).

Wisconsin is one of the most studied cases in redistricting litigation. The state’s Republican-drawn 2011 map was challenged in *Gill v. Whitford* (2018), where the Supreme Court dismissed the case on standing grounds without ruling on the merits of partisan gerrymandering.

Wisconsin’s political geography is inherently challenging for Democrats: urban voters (Milwaukee, Madison) are heavily concentrated, naturally wasting many Democratic votes even in neutral maps. Our ensemble confirms this—the spectral baseline itself produces $EG = +0.012$ (slightly Republican-favoring), a consequence of geography not manipulation. But the enacted map achieves $EG = -0.246$, a large *Democratic* swing relative to the spectral baseline, and a cut-edge ratio (0.042) below even the spectral plan (0.036), placing it deep in the ensemble tails.

We additionally measured **MST diameter**: the enacted map scores 31.50 vs. an ensemble mean of 35.40 ($\sigma = 2.30$), placing it at the 3.6th percentile ($p = 0.074$). Like PA, Wisconsin’s districts are directionally more compact on this measure, but the result falls just outside conventional significance. The partisan metrics remain overwhelming. Composite severity: **3.43**. **Conclusion: Wisconsin’s high severity score reflects a map that is a statistical outlier even after accounting for the state’s natural geographic partisan lean.**

5.5 Cross-State Comparison

State	k	severity score	enacted EG	enacted cut ratio	metrics
Wisconsin	8	3.43	−0.246	0.042	5 (incl. MST)
Pennsylvania	18	3.40	−0.026	0.055	5 (incl. MST)
North Carolina	13	2.28	−0.054	0.082	5 (incl. MST)
Maryland	8	2.19	+0.200	0.189	5 (incl. MST)

Table 1: Cross-state composite severity ranking (real MGGG data, 900-plan ensembles, all states re-run with 5 metrics including MST diameter). Severity is the average of $-\log_{10}(p)$ across metrics; higher = more extreme outlier. EG sign: negative = more Democratic waste (Republican-favoring map), positive = more Republican waste (Democratic-favoring map).

All four states were re-run with 5 metrics (including MST diameter). Wisconsin (3.43) and Pennsylvania (3.40) are the strongest outliers—within a rounding error of each other. For Wisconsin, all partisan metrics remain at $p < 0.001$; MST is directionally compact ($p = 0.074$) but just misses significance, pulling the severity down from the naive 4-metric estimate. Pennsylvania’s map is a strong outlier in the *good* direction across all 5 metrics.

North Carolina (2.28) and Maryland (2.19) rank lower. MST diameter is significant for both ($p = 0.007$ and $p < 0.001$, respectively)—both enacted maps are more compact on graph-theoretic diameter than a neutral random redistricting. For Maryland this is striking: the map is shape-compact despite its notoriety, but the partisan manipulation is severe. For North Carolina, compactness (including MST) is the dominant signal; partisan metrics do not reach significance.

The key insight: *no single metric is decisive*. The strength of the ensemble test is that it asks whether a map is extreme on multiple independent measures simultaneously. A map that is an outlier on compactness, efficiency gap, and mean–median—all at once—is much harder to explain away as geographic accident.

6 Limitations

MCMC mixes slowly. Our single-flip sampler changes one precinct at a time. This means consecutive maps in the chain are very similar, and the chain takes many steps to explore the full space of valid maps. We run 3 independent chains and report $\hat{R}$ and ESS to flag cases where mixing is poor. For states with complex geography, a recombination sampler (which merges and re-splits entire districts) would mix faster [1].

Population data is approximate. We use 2010 Census population counts. By 2016, populations had shifted. Precincts that were balanced in 2010 may be over- or under-populated by 2016. A fully rigorous analysis would use more recent estimates.

We only measure five metrics. Gerrymandering can be detected through many other lenses (racial composition, incumbent protection, etc.) that we do not measure here.

The enacted plan is the only hypothesis tested. We compare one enacted map against the ensemble. We do not test whether a “better” gerrymandering could be even more extreme and still pass our tests.

7 Graph Theory and Networks Concepts Used

This project applies several ideas from the NETS 1500 course:

Graph partitioning. The core problem—dividing a graph into k connected, balanced subgraphs—is a fundamental graph problem. We use spectral methods (Fiedler vector) rather than the brute-force approach.

Spectral graph theory. The Laplacian matrix $L = D - A$ and its eigenvectors encode the connectivity structure of the graph. The second eigenvector (Fiedler vector) reveals the slowest bottleneck in the graph and provides a natural partition.

Community detection. Newman–Girvan modularity is the standard community detection objective function. We apply it as a compactness metric: a partition that respects communities is compact; one that breaks them up is not.

Random walks on graphs. Our MCMC sampler is a random walk on the *space of valid partitions*. Each step changes one node’s district membership. The stationary distribution of this walk approximates a uniform sample over valid plans.

Network science and social networks. Precinct adjacency graphs have strong community structure—politically similar precincts cluster geographically, just like homophily in social networks. The cut ratio (a conductance-like measure) and modularity directly apply community-detection ideas from network science.

8 Conclusion

The Verdicts: Which States Are Gerrymandered?

Here is what the data actually says, state by state.

Wisconsin: Gerrymandered (strongest evidence). Wisconsin’s 2011 Republican-drawn map is a statistical outlier on *every partisan metric*—efficiency gap and mean–median—both at $p < 0.001$. Compactness metrics (cut-edge ratio, modularity) also reach $p < 0.001$. The MST diameter sits at the 3.6th percentile ($p = 0.074$), directionally compact but just outside significance. That means out of 900 randomly drawn maps, not one comes close to matching the enacted map’s combination of compact shape and extreme partisan skew.

The most telling number is the efficiency gap swing: a neutral map of Wisconsin (spectral baseline) produces $EG = +0.012$, reflecting a slight natural Republican geographic advantage. The enacted map achieves $EG = -0.246$. That is a swing of 25 percentage points in the direction of wasting Democratic votes. To put that in concrete terms: if you drew 900 maps at random and picked the worst one for Democrats, you still would not get close to the enacted map. **Our verdict: gerrymandered.** This aligns with the legal case *Gill v. Whitford* (2018), where the court acknowledged the map’s extreme nature but declined to rule on the merits.

Maryland: Gerrymandered—but you would not know it by looking. Maryland’s 2011 Democratic-drawn map has a cut-edge ratio of 0.189 and modularity of 0.682. These compactness numbers are not statistically unusual— $p = 0.429$ and $p = 0.450$, both well within normal range. The districts do not look bizarrely shaped on a map. And the MST diameter (22.50, 0th percentile, $p < 0.001$) is actually significantly *more compact* than a random redistricting—the district network is geometrically tight even as individual shapes look irregular.

But look at the partisan numbers: efficiency gap = +0.200 ($p < 0.001$) and mean–median = -0.013 ($p = 0.006$). Both are highly significant. The enacted map systematically wastes Republican votes at a rate no neutral random map would produce by accident. Maryland’s 6th Congressional District—redrawn in 2011 to flip from safe Republican to safe Democratic by absorbing Washington D.C. suburbs—is the smoking gun. The shapes look fine; the politics are rigged.

This is what political scientists call a **stealth gerrymander**: you cannot detect it just by looking at the map. You need the partisan metrics. **Our verdict: gerrymandered** (Democratic partisan gerrymander, confirmed by partisan metrics alone).

North Carolina: Detected as unusual, but mostly on compactness. North Carolina’s 2016 congressional map scores 2.28 on our severity index and is flagged as a compactness outlier—but in the *good* direction. The enacted map’s cut-edge ratio (0.082) sits at the 0th percentile of the ensemble (mean 0.214), meaning it is *more compact* than all 900 randomly drawn plans. The MST diameter (20.15) reinforces this: at the 0.6th percentile ($p = 0.007$), the districts are significantly tighter on graph-theoretic diameter than a neutral map. This is counterintuitive: a famously gerrymandered state whose enacted map is *more compact* than random chance?

The partisan metrics do not reach significance: efficiency gap $p = 0.283$, mean–median $p = 0.222$. This does not mean the map is fair; it means our 900-plan ensemble does not have enough statistical power to isolate the partisan manipulation from natural geographic variation given North Carolina’s competitive two-party split. North Carolina demonstrates a key limitation: *ensemble methods need large enough samples and the right baseline to detect every form of manipulation*. **Our verdict: evidence of compactness manipulation (districts more compact than chance), but partisan gerrymandering not conclusively confirmed by our metrics alone.**

Pennsylvania: The Control Case—What a Fair Map Looks Like. Pennsylvania’s 2018 remedial map, drawn by a court-appointed special master with fairness as an explicit goal, scores 3.40 on our 5-metric severity index. But unlike Wisconsin, it is not an outlier in a bad way.

The enacted map’s cut-edge ratio (0.055) is lower than all 900 random plans (ensemble mean 0.116). Its efficiency gap (-0.026) is far closer to zero than the ensemble average (-0.134), correcting for the state’s natural Republican geographic lean. The MST diameter (24.9) sits at the 2.8th percentile ($p = 0.096$), directionally compact though just outside strict significance. Every metric places the enacted plan deep in the tails—but in the tails corresponding to *fairer and more compact* maps. **Our verdict: not gerrymandered.** Pennsylvania is the proof of concept: when courts intervene with an explicit fairness mandate, the statistical signal is clear and in the opposite direction from a gerrymander.

What the Numbers Actually Mean

A lot of redistricting debate gets lost in legal language or vague accusations. Our method puts specific numbers on specific claims:

- An efficiency gap of ± 0.08 (8 percentage points) is generally considered the threshold for a problematic gerrymander in the academic literature [2]. Wisconsin’s -0.246 is more than three times that threshold.
- Maryland’s $+0.200$ is similarly extreme—one of the most Democratic-skewed congressional maps in the country.
- Pennsylvania’s -0.026 is nearly perfectly symmetric, a direct result of court-mandated fairness constraints.
- A p-value below 0.05 means the enacted map falls outside the 95th percentile of what random chance would produce. Every significant result in our study is at $p < 0.01$ —far more stringent than that standard.

The Big Lesson: You Need Both Metrics

Maryland proves that *you cannot detect a gerrymander just by looking at district shapes*. The shapes are fine; the partisanship is not. Wisconsin proves the reverse is also dangerous: even when districts look compact on a map, the partisan outcome can be extreme. You need to measure both compactness and partisan fairness, compare them to a realistic ensemble of neutral maps, and look for outliers across multiple metrics simultaneously.

Our ensemble approach—generate hundreds of neutral maps using a principled random walk, then ask where the real map falls—is the standard method used by expert witnesses in actual redistricting litigation. This project implements that framework from scratch, using only precinct shapefiles and election returns available for free from the MGGG research group.

The full pipeline, including all algorithms and 61 unit tests, is implemented in the `gerrydetect` Python package. To reproduce:

```
python scripts/download_mggg_states.py all
python scripts/run_real_all_states.py
```

References

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